

- There are 5 Questions for 20 marks.
- Write your answers neatly, legibly, and logically. Give supporting arguments for your answers. Keep your answers thorough and concise. Your arguments must follow from the basics discussed in class. You may use any result from class directly (without re-deriving).
- Problems marked *Optional* do not count towards the total for this quiz, but your solutions to these problems may be considered while assigning a letter grade at the end of the course.
- Good luck !

1. [5] (a) Let the function $f : \mathbb{C} \mapsto \mathbb{C}$ be defined as $f(z) = \bar{z}$, where $\bar{z}$ is the complex conjugate of z . Is f a linear function ?
- (b) The function $\varphi : \mathbb{R}^3 \rightarrow \mathbb{R}$ satisfies

$$\varphi(2, 0, 0) = -1, \quad \varphi(-1, 1, 1) = 1, \quad \varphi(0, 1, 1) = 1/2.$$

Choose one of the following, and justify your choice:

- (i) φ must be linear; (ii) φ could be linear; (iii) φ cannot be linear.

2. [3] Define the following function $f : \mathbb{R}^n \mapsto \mathbb{R}$

$$f(\underline{x}) = \text{number of non-zero entries in } \underline{x}.$$

Which of the requirements for a function to be a norm are satisfied by f ? Is f a norm ?

3. [6] (a) Among all points on the (3-d) sphere $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$, which point has the largest value of $x + 2y + 3z$? *Hint: Use the Cauchy-Schwarz inequality.*
- (b) Show by using the Cauchy-Schwarz inequality, that if a_j and b_j are nonnegative real numbers and satisfy

$$1 \leq a_j b_j \quad \text{for all } j = 1, 2, \dots, n,$$

then one also has

$$1 \leq \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right) \left(\frac{b_1 + b_2 + \dots + b_n}{n} \right),$$

i.e. the product of averages is also at least 1.

4. [3] Consider a game played by two players with a single pile of matches. The players take turns alternately, with Player 1 going first. On each turn, a player must remove either one or two matches from the pile. The game ends when the pile is empty; the player who takes the last match wins.

Assume that the pile initially contains $3n$ matches for some $n \in \mathbb{N}$. Both players know the initial number of matches and all previous turns.

Suppose you are Player 2. What winning strategy can you adopt? Prove that your strategy works.

Hint: Like in the tutorial problem, you may find it useful to choose a strategy that ensures that after both players have completed their turns, the game reduces to a smaller instance of the same form.

Optional: Show that if the pile initially contains $3n + 1$ or $3n + 2$ matches, then Player 1 has a winning strategy.

5. [3] A group of 5 people are represented as nodes in the graph below. At each time $t = 0, 1, 2, \dots$, each person i has a message v_{it} (the messages are all real numbers). At time $(t + 1)$, the messages are updated according to the rule

$$v_{i(t+1)} = \sum_{j \sim i} v_{jt}, \quad \text{for } i = 1, 2, \dots, 5;$$

where $j \sim i$ means that j and i are connected by an edge in the given graph. Let $\underline{v}_t = (v_{1t} \ v_{2t} \ \dots \ v_{5t})^T$ be the vector of messages at time t .

3n-1
3n-2
3n-1
3n-1

- (a) Express $\underline{v}_{t+1}$ in terms of $\underline{v}_t$ as $\underline{v}_{t+1} = A\underline{v}_t$ where A is a 5×5 real matrix. Give an interpretation for the entries of A .
- (b) Suppose vector of initial messages (at time $t = 0$) is $\underline{v}_0$. Is $\underline{v}_4$ a linear function of $\underline{v}_0$? If so, give a corresponding matrix representation.
- (c) Give an interpretation for the entries of A^2 .
- (d) *Optional:* Generalize the interpretation above to entries of A^n for $n = 1, 2, 3, \dots$, and verify by induction.

